American International University - Bangladesh (AIUB)

Programming in Python [A]

Mid term project

Title: Metro Rail Management System

Course Title	Programming in Python [A]
Project Type	Midterm Project
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Github link	https://github.com/Grontho-Roy-Nirob/Python-Project-Midterm
Submitted To	**MD. TANZEEM RAHAT**

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1. Introduction

Urban metro transport requires quick ticket processing, transparent fare calculation, and reliable passenger record handling. Manual ticket handling is time consuming and more prone to errors. To address this issue, the team developed a Python based Metro Rail Management System that automates ticket booking and management activities through an interactive graphical user interface.

2. Tools and Technologies

Component	Used In Project	Purpose
Python	Core language	Implements business logic and application flow
Tkinter	GUI toolkit	Builds forms, buttons, messages, and ticket table
ttk	Tkinter themed widgets	Provides combobox, treeview, and scrollbar
tkcalendar	DateEntry widget	Supports filtering tickets by date
JSON	Local data storage	Saves and reloads ticket records
Pillow	Image handling	Displays the metro image on the welcome page

3. System Overview

The project contains a welcome screen and a main management window. The welcome window shows the project title and an image, then opens the main ticketing module. The main window is divided into two sections: the left panel receives passenger input and operation commands, and the right panel displays all ticket records in tabular form.

4. Functional Modules

Module	Description	Key Outcome
Ticket class	Stores passenger name, stations, fare, quantity, total cost, type, date, and unique ticket ID.	Creates structured ticket records.
MetroSystem class	Manages fare calculation, booking, searching, deleting, saving, loading, and total handling.	Controls the main business logic.

MetroApp class	Builds the Tkinter interface and connects button actions to system methods.	Provides user interaction.
Data persistence	Reads and writes ticket information in tickets.json.	Preserves records between sessions.
Validation	Checks empty fields, invalid quantity, same start and end station, and numeric characters in names.	Improves reliability and input accuracy.

5. Features of the System

Feature 1: Ticket Booking

This feature allows users to book metro tickets by entering passenger name, start station, destination station, quantity, and passenger type. After submitting the information, the system automatically calculates the fare, generates a unique ticket ID, and stores the ticket record. This makes the booking process faster, easier, and more accurate.

Feature 2: Automatic Fare Calculation

The system automatically calculates ticket fare based on the distance between stations and passenger category. A base fare and per station charge are applied, and discounts are given for student and disabled passengers. This feature ensures fair and transparent fare processing without manual calculation.

$$\text{Fare} = \text{Base Fare} + (\text{Distance} \times \text{Per Station Fare})$$

- Base fare = 20 BDT
- Per station fare = 10 BDT
- Student passenger discount = 50%
- Disabled passenger discount = 80% of the base amount is reduced, leaving 20% payable as coded in the program
- If start and end stations are the same, fare becomes 0 and ticket creation is blocked by validation

Feature 3: Input Validation

The system checks whether the entered information is valid before creating a ticket. It prevents empty fields, invalid quantities, same start and end stations, and incorrect passenger names containing numbers. This feature improves reliability and reduces user errors.

Feature 4: Ticket Record Display

All booked tickets are displayed in a table using the Treeview widget. The table shows ticket ID,

passenger name, route, fare, quantity, passenger type, total amount, and booking date. This helps users and operators easily monitor all ticket records in one place.

Feature 5: Search Ticket

The application provides a search option to find a ticket quickly using ticket information. This feature saves time and helps users locate specific booking records without checking the full list manually.

Feature 6: Delete Ticket

Users can delete a selected ticket record from the system. After deletion, the ticket is removed from both the table and the stored file. This feature helps maintain updated and error-free records.

Feature 7: Date-wise Ticket Filter

The system can filter ticket records according to a selected booking date. This allows users to view only the tickets booked on a specific day. It is useful for daily record analysis and management.

Feature 8: Data Storage using JSON

Ticket information is saved in a JSON file so that records remain even after closing the application. When the program starts again, the saved data is automatically loaded. This feature provides simple and effective data persistence.

6. Sample Output Interpretation

The sample data file included in the project demonstrates that the system successfully stores ticket IDs, passenger names, route information, fare, quantity, passenger type, total amount, and booking date. This confirms correct integration of booking, discount handling, and persistent storage.

Example Case	Route	Passenger Type	Observed Result
Case 1	Uttara North to Pallabi	Student	Discounted fare stored correctly
Case 2	Uttara North to Mirpur 10	Disabled	Large discount applied and record saved
Case 3	Mirpur 11 to Uttara Center	General, quantity 2	Total amount calculated using quantity

7. workflow

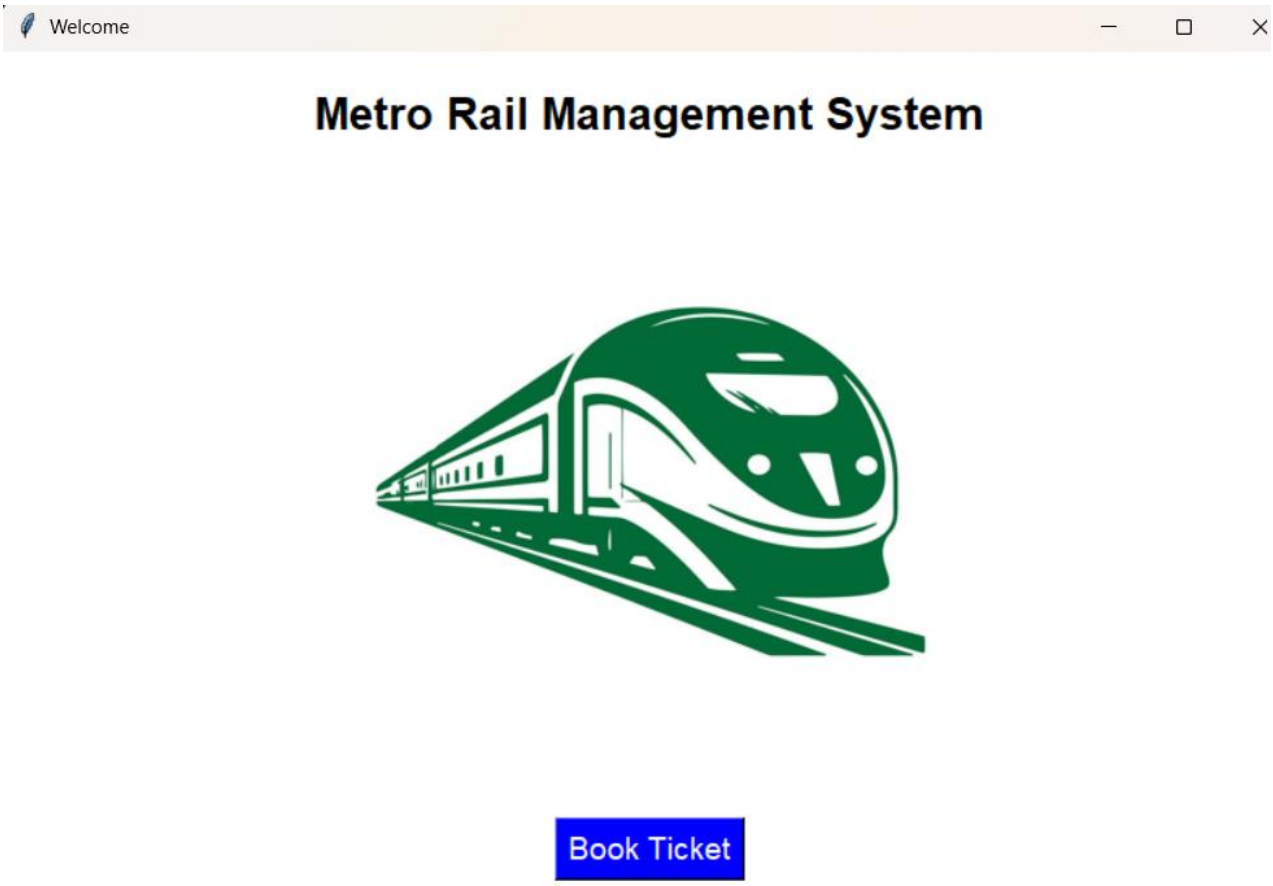

Fig -2 welcome page

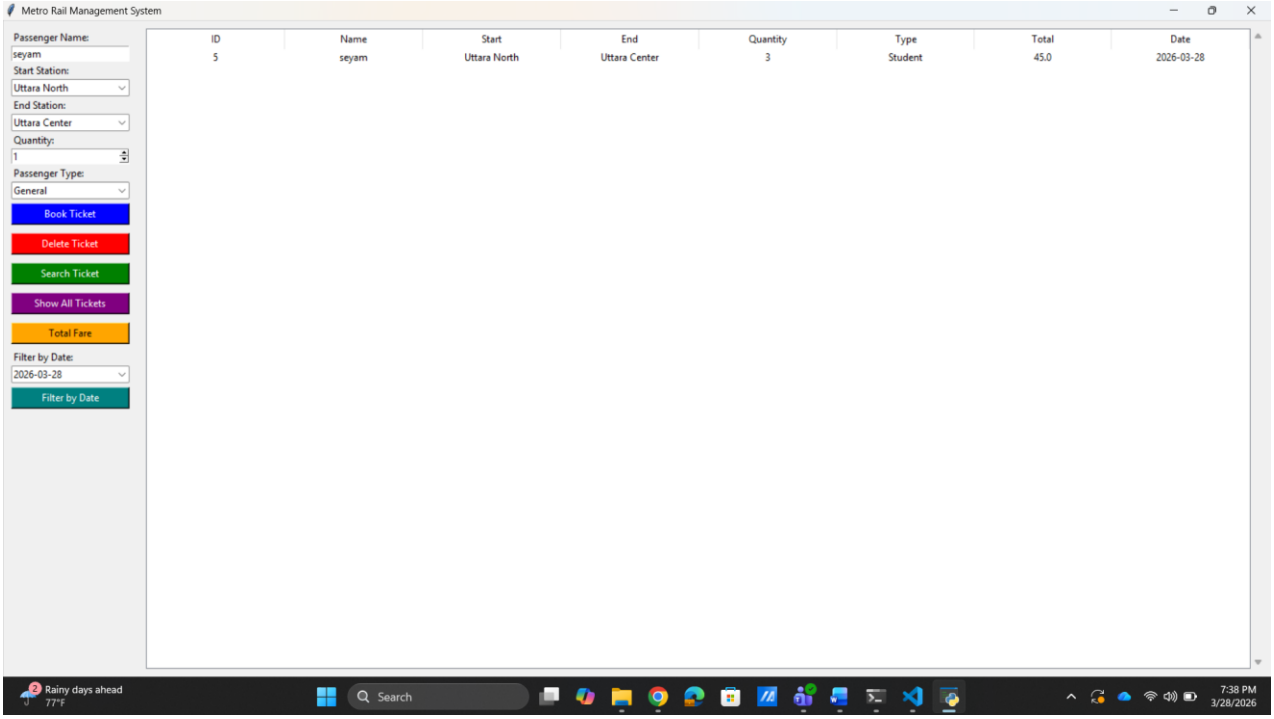

Fig -3 Homepage

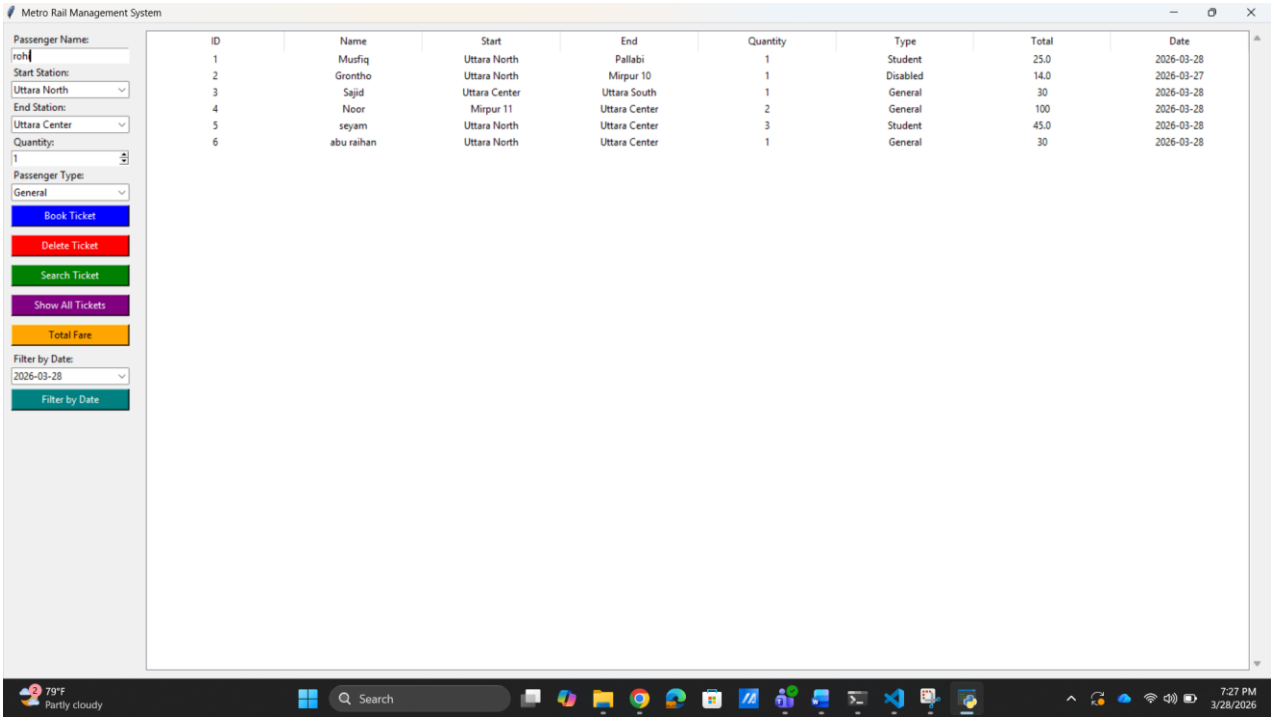

Fig-4 insert name

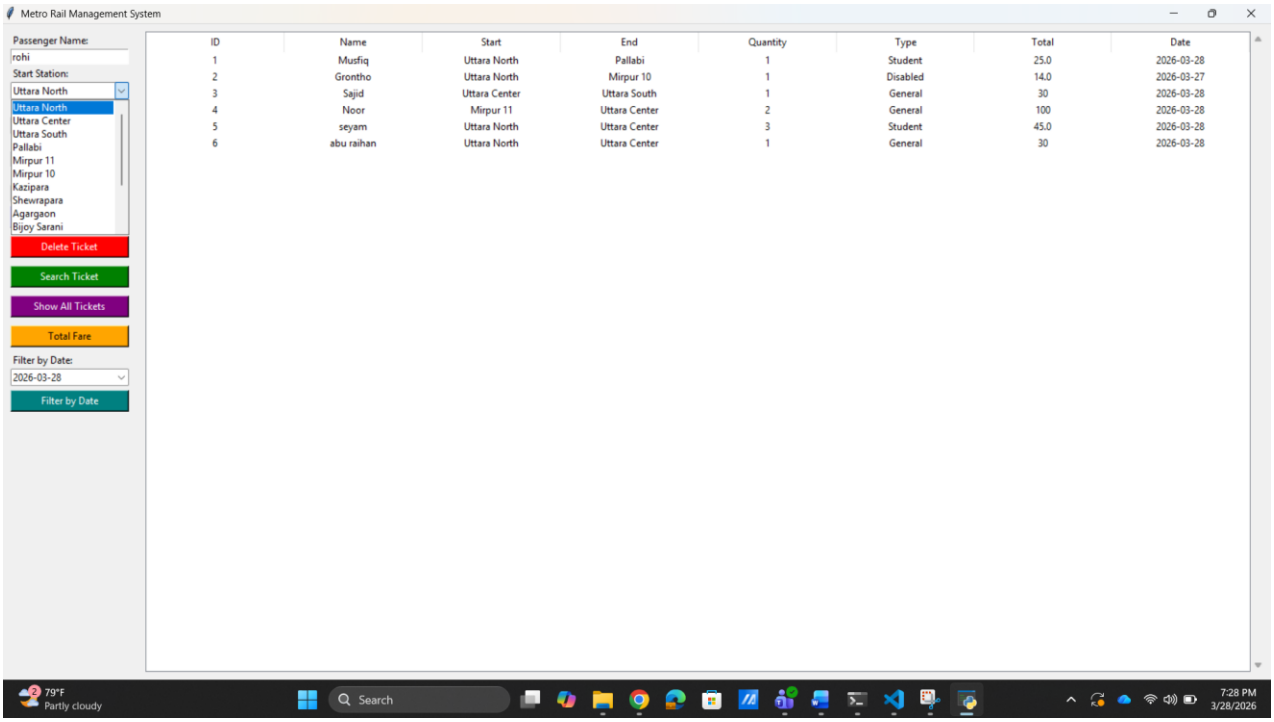

Fig -5 select station

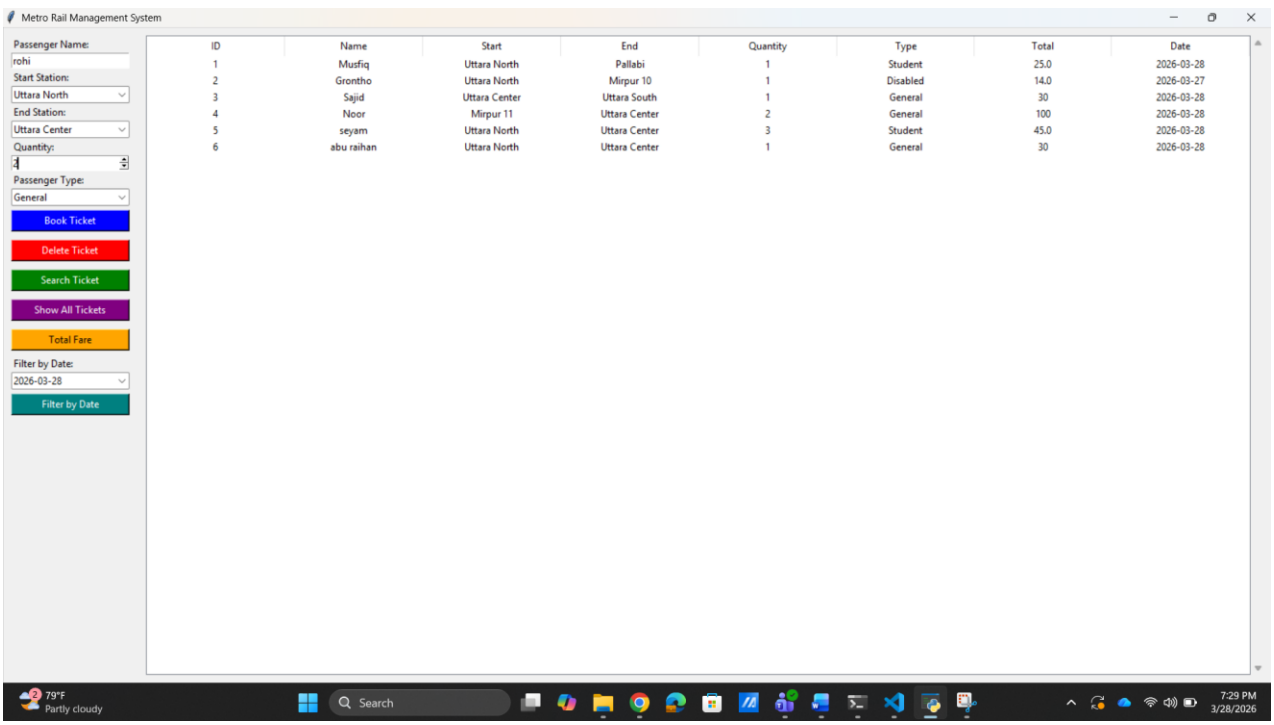

Fig-6 Quantity

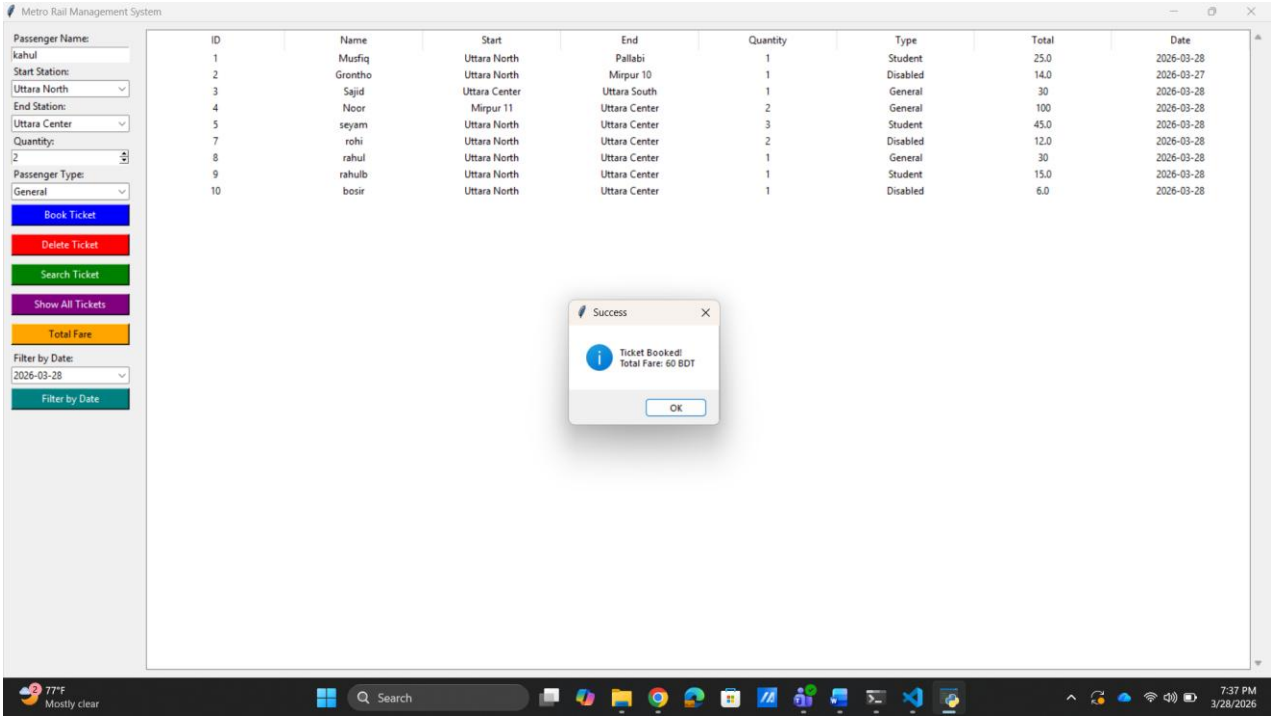

Fig-7 Booked ticket

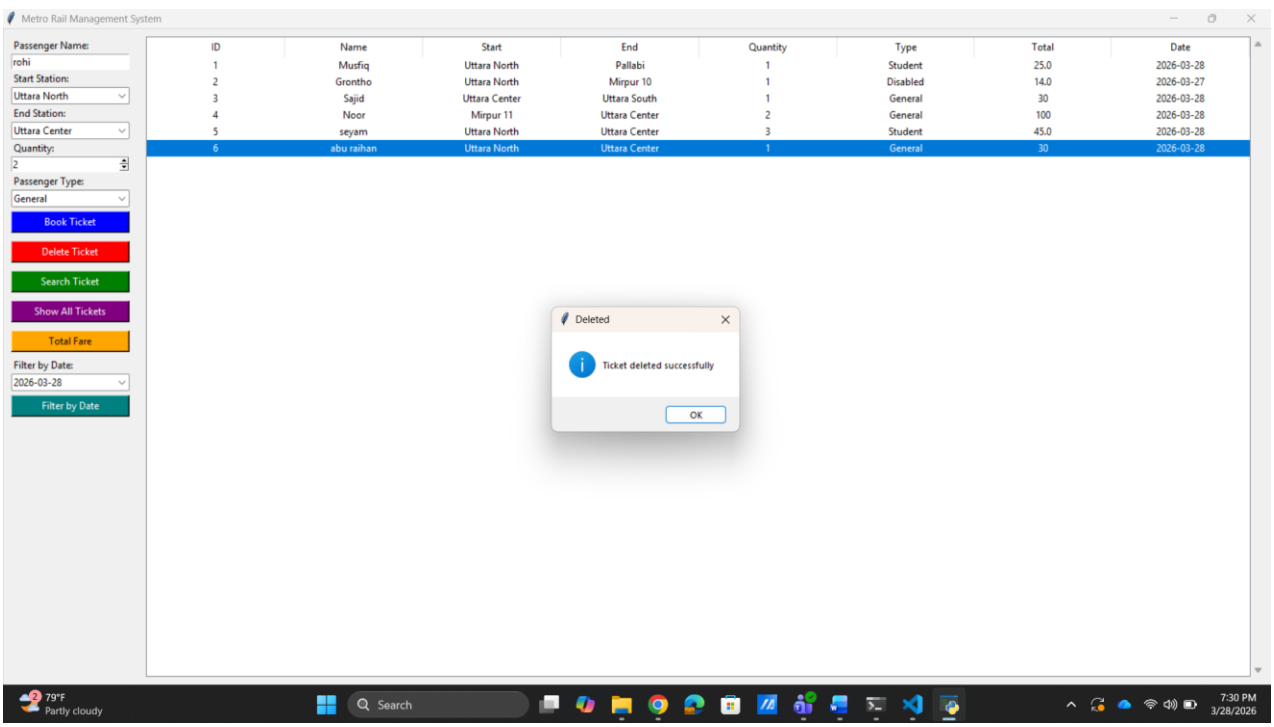

Fig –8 deleted ticket

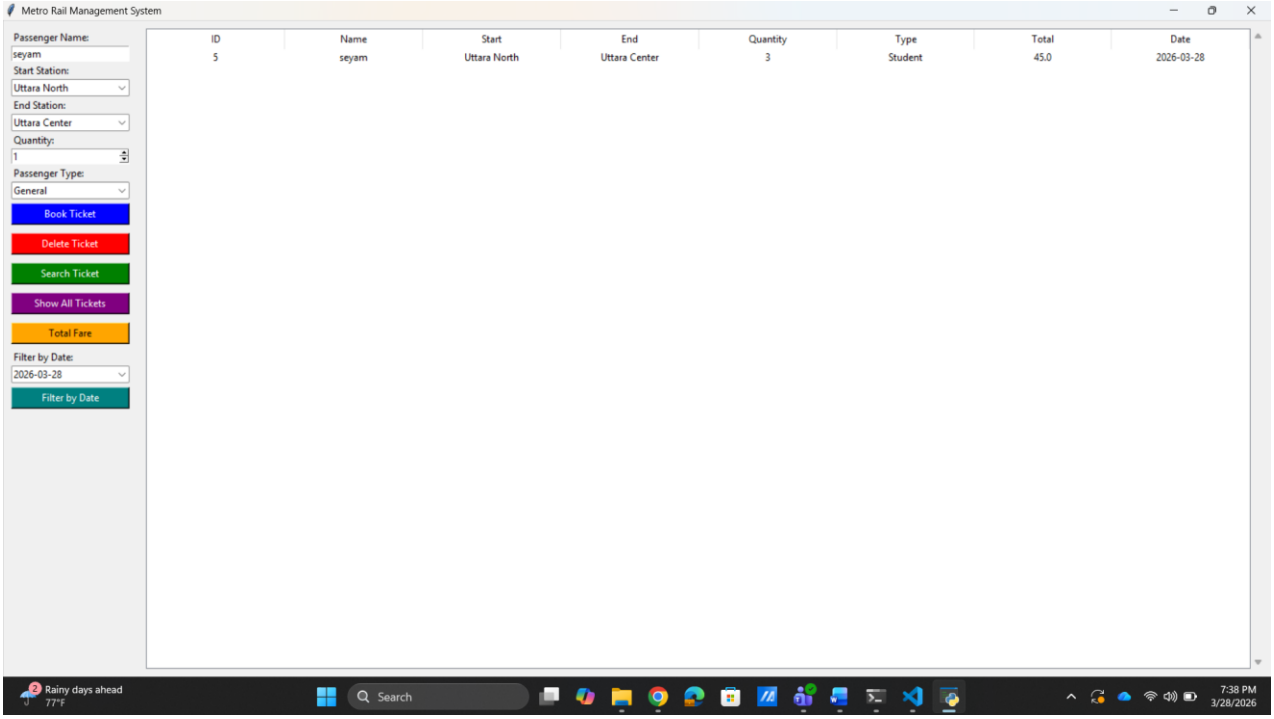

Fig-9 Search Ticket

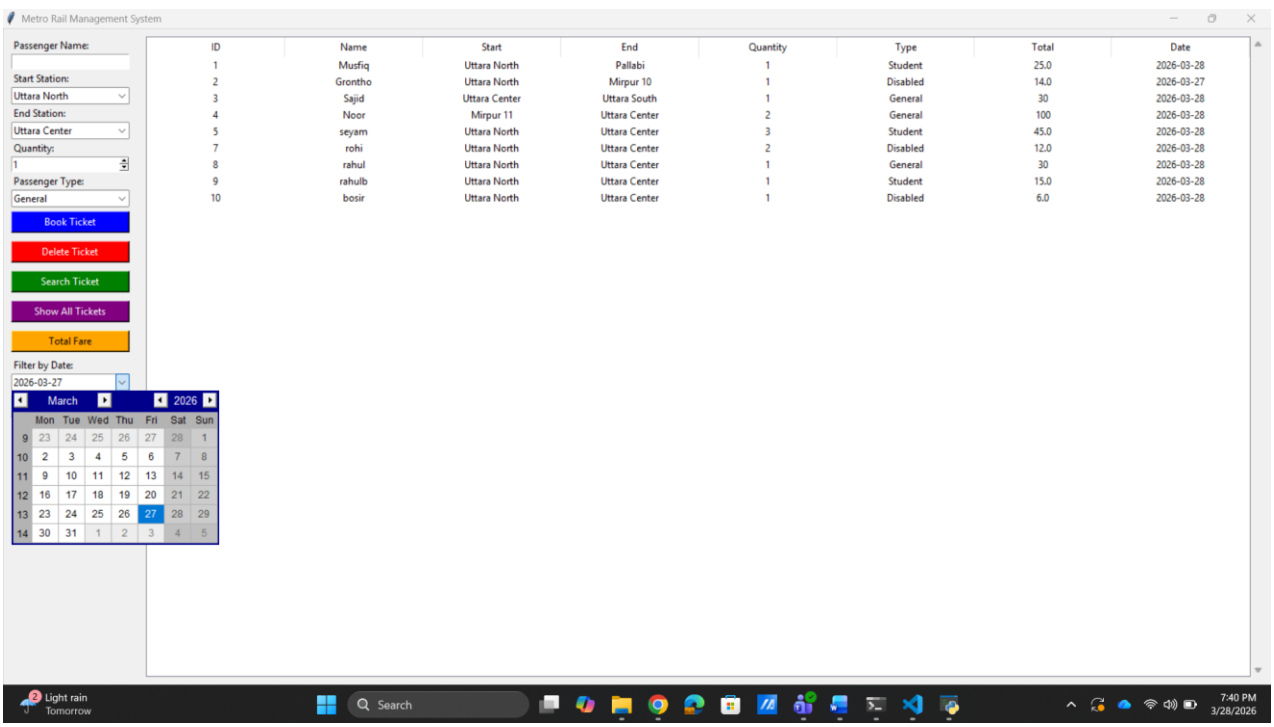

Fig-10 -Date wise ticket booked

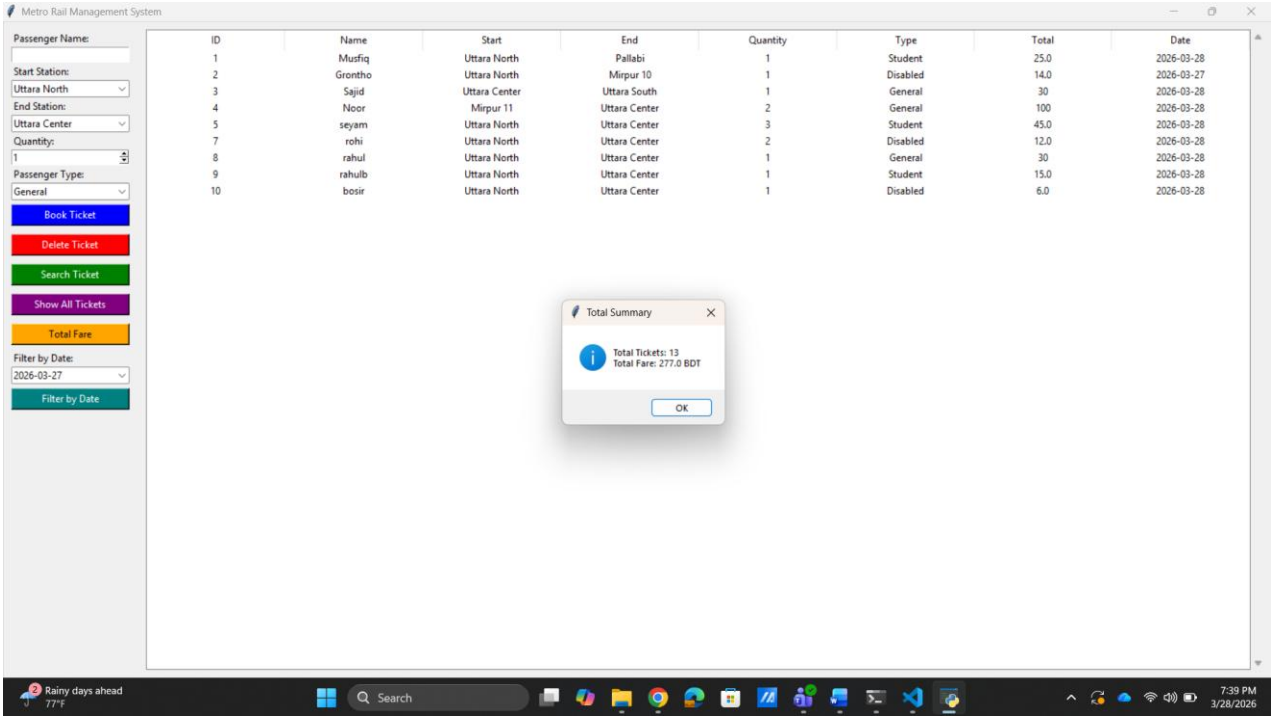

Fig 11- Total Fare and ticket quantity count

8.How to run

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Open the project in VS code .Download the requirement.txt using this command-
(python -m pip install -r requirements.txt)

then open the welcome.py and run this project.
```

9. Conclusion

The Metro Rail Management System fulfills the main objective of automating a basic metro ticket management process using Python. The project successfully combines GUI development, object oriented design, data validation, discount based fare calculation, and JSON file handling in one complete application.